

THE LITTLE Server That Could

Thanks to virtualisation, a single server can manage a lot—including saving money and managing resources better

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In an ideal world, you'd invest in one server for each function in your office—one file server, one domain controller, one proxy server, and so on. All are isolated, so changes in one won't affect the others, and if one crashes, the rest remain unharmed. This is all nice, but did you know that even a Pentium II running on Linux (or a Pentium III running Windows Server 2003) can act as a file server for a hundred-odd people? Even entry-level servers have much more power; imagine the processing power you waste if you buy one dedicated server for each function!

The best way to put all this unused power to use is *server virtualisation*. Despite it being around for ages, very few people are even aware of the concept; however, it's been gathering momentum over the past couple of years.

What It Is

It's a broad term and is used in many contexts, but the simplest explanation is that virtualisation splits a single server into many "virtual" servers, each running a different operating system. For example, a dual-core processor-based server can be virtualised to behave like two single core-based servers, and so on. If you run your file server on one, you've got plenty of processing power to run your proxy server on the other. It is estimated that it will take eight such virtual servers to bring CPU utilisation up to 50 per cent at full load, as opposed to the paltry 5 for just one OS. And since you paid good money for that CPU, you might as well utilise it properly.

How It Works

When you install a single OS, it talks directly to the hardware. Any application that needs to perform a system task—writing to the hard disk, for example—needs to do so through the OS. If many applications need access to the hardware, the OS lets them have it on something of a "time-share" basis. With virtualisation, this concept goes one step ahead. Instead of the OS, you now have a *hypervisor* or Virtual Machine Monitor (VMM) running directly on the hardware, and the operating system runs on top of the hypervisor. Another way virtualisation is implemented is with the hypervisor running within a "host" operating system, and the remaining "guest" operating systems running on that hypervisor. In the first case, one layer of code has been added between applications in guest operating systems and the hardware; in the second, two.

Of course, you're probably appalled at the performance hit this represents. It's been estimated that for every layer of code between an application and the hardware, performance drops by a factor of ten. Virtualisation software developers have been battling this problem for a long time—at the stage they've reached today, the performance hit is a fair price when you consider the bigger picture (which we'll come to in a bit). Intel and AMD have started supporting virtualisation on their new server processors, so we should see better performance from virtualised servers in the future.

Why Do I Care?

The first advantage of virtualisation, as mentioned before, is the consolidation of many functions into one physical server. This saves you money and ensures that your CPU is utilised better—but there's more. If you're running a file server, proxy server and e-mail server on the same server, and that server gets infected with a virus, three of your office's most important IT services go down. If you've got each service run-

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